

Predict, test, explain.

Learning to build connected data tracking across watchOS and iOS — by writing down what I expected **before** every experiment, and keeping every prediction that turned out wrong.

Growth focus statement

Learn how connected data tracking works across watchOS and iOS – background execution, data ownership, transport reliability, source of truth – by producing one complete workout captured on the watch and reviewed on the phone, **together with a log of what I predicted wrongly**, so that I can build a data crossing that holds through screen-off and force-quit, and explain why it holds.

**The log of wrong predictions
was in the plan from day one.**

I train HYROX: eight runs, eight stations, and the rest between them counts. I wanted an app that records the race on the watch and explains it honestly on the phone.

Zero.

HealthKit, workout sessions, background execution, moving data between two devices — I had used **none** of them.

So I did not plan to build first. I planned to **predict** first. Before every experiment I wrote what I expected; afterwards, what happened, and the gap between them.

Rule, stated at the top of the workbook: a prediction is never edited once the test has run. A wrong prediction is the evidence, not a mistake to hide. Git history shows each prediction committed before its result.

6 CYCLES	24 EXPERIMENTS	101 COMMITTS	2,172 WORKBOOK LINES
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Every experiment ran on an Apple Watch Ultra 3 and an iPhone 16 Pro Max. Never a simulator.

The largest planned task produced zero code.

Cycle 4 was scheduled as the biggest build in the project: a connection to move data from the watch to the phone. I built none of it.

Cycle 3 measured it first

A finished workout reached the phone in under 14 s – phone locked, in another room.

Cycle 4 confirmed the limit

A full-length race – 25 segments, 601 characters – crossed complete. No truncation.

So it became a decision

A written decision record, with the conditions that would reverse it. Not an omission.

It is the result I am most confident about, because I only knew it was safe to delete after measuring it.

A rest of 2.03 seconds.

In Cycle 5 I stopped tapping through tests at a desk and ran a **real session**. Out of breath, pressing a button on my wrist between stations. At the second rest, I pressed it twice by accident.

24.76_s

ROXZONE 1

2.03_s

ROXZONE 2 — IMPOSSIBLE

INTACT

WHAT MY CHECKER REPORTED

**Every check I had written
said the data was fine.**

Timestamps in order. Nothing negative. Everything inside the workout. The check tested completeness, and completeness was never the question.

10.58



Review

WORKOUT REVIEW

Tuesday, 1 September 2026
at 10.51

MEASURED

TOTAL DURATION

5m 12s

MEASURED

ACTIVE ENERGY

45 kcal

MEASURED

Segments

Run 1

START OFFSET

+42s

DURATION

47s

10.58



Review

MARKED

DERIVED

FROM TWO MARKED TIMES

Station 2 — Sled Push

START OFFSET

+4m 34s

MARKED

DURATION

38s

DERIVED

FROM MARKED
+ MEASURED

Totals

Run

2m 17s

DERIVED

FROM MARKED TIMES

Roxzone

1m 0s

DERIVED

A green signal hiding a failure. Seventeen times.

THE SIGNAL SAID

WHAT WAS ACTUALLY TRUE

Running normally

79.4% of elapsed time was not being counted. No crash, no error.

Authorised

Read permission was denied. HealthKit returns an empty set, never an error.

Build succeeded

Missing usage description. Instant termination, before any catch could run.

INTACT

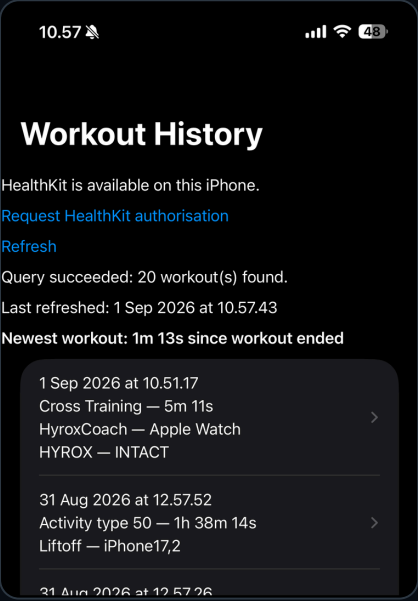
Every offset was negative. The check tested completeness, never plausibility.

AGREE

Segment names matched. **No durations were compared**, and nothing said so.

MEASURED

Apple’s own metadata: HKWeatherHumidity 4900 %.



My own instrument · 1 September

Query succeeded: 20 workout(s) found. Not one of them was mine. I concluded HealthKit had deleted them – **and they came back the same day, unchanged.** The retraction is still in the workbook, above the wrong text.

All seventeen are indexed with workbook line numbers. Five were produced by instruments built in this project to catch exactly this. Two were my own reasoning errors, both retracted in place, with the wrong text kept above the retraction.

A passing test only proves

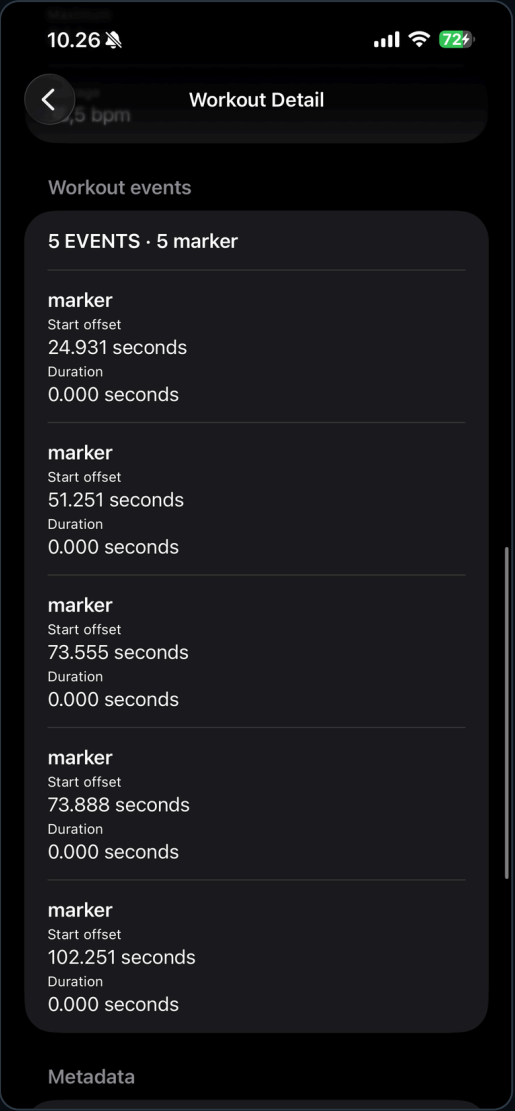
0.333 seconds apart. Recorded without complaint.

I pressed the lap button in Apple's own Workout app twice, as fast as I could. It stored both marks. No debounce, no flag, nothing indicating the second was suspect — the same failure as my 2.03-second roxzone.

Apple has not solved this either.

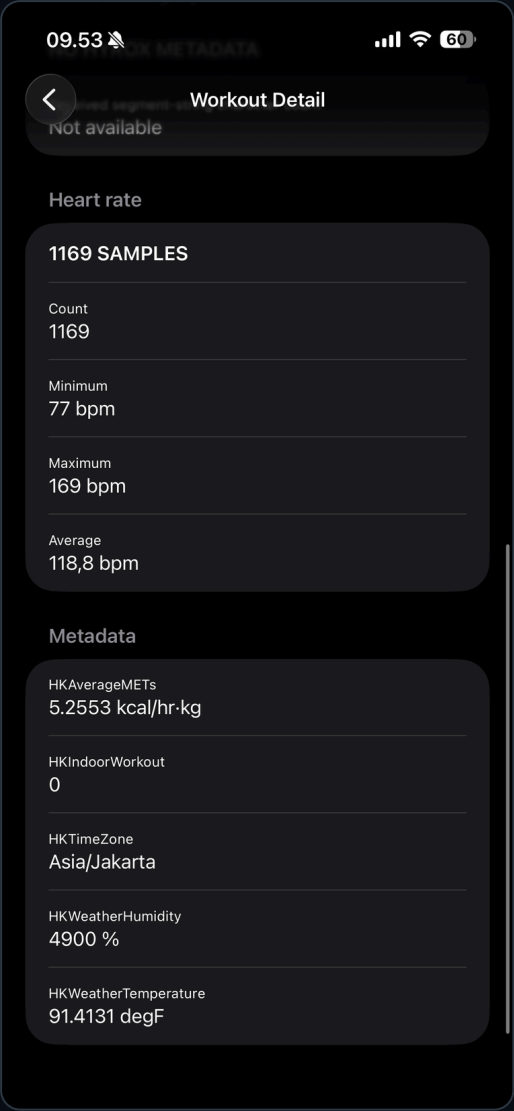
So the problem I hit is not inexperience. Marking boundaries accurately mid-effort is unsolved on the platform, and solving it properly is a real contribution rather than catching up.

Not claims. Screens.



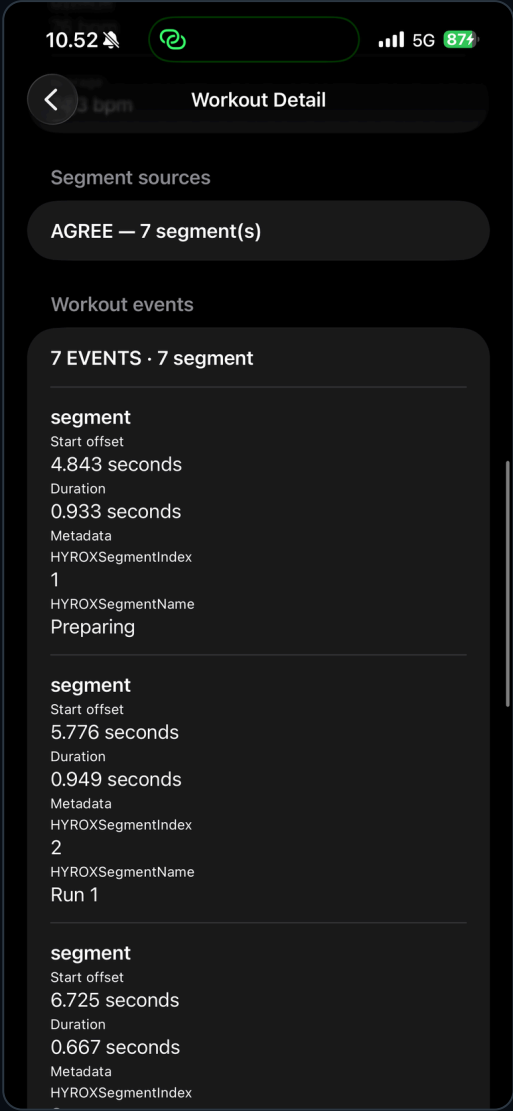
E6.0 • APPLE’S OWN APP

Five markers, every duration 0.000.
Two of them at 73.555 and 73.888 – **0.333 seconds apart, both kept, neither flagged.**



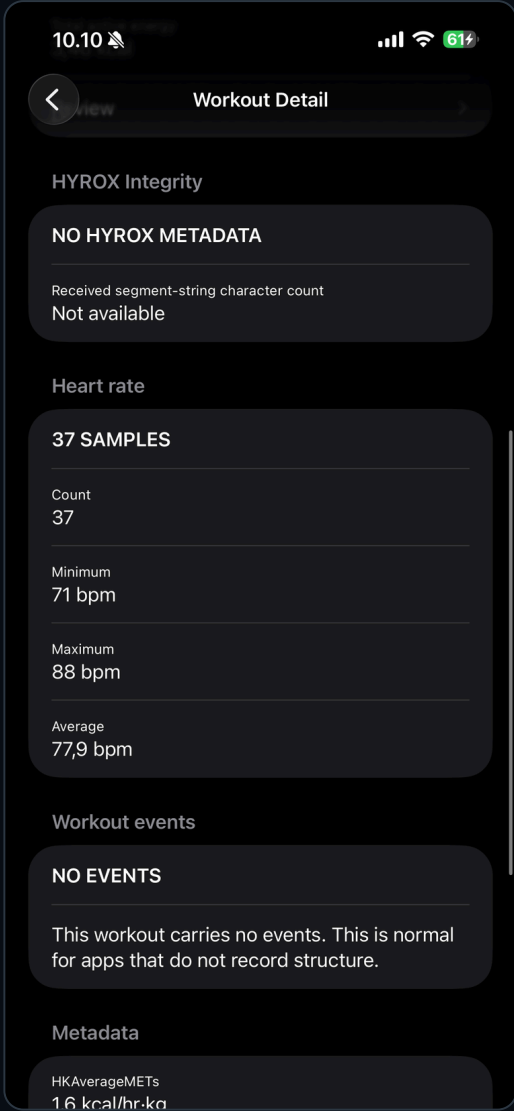
E5.1 • MEASURED IS NOT SAFE

Apple’s own metadata:
HKWeatherHumidity 4900 %.
Complete, well-formed, plausible to any checker, and false.



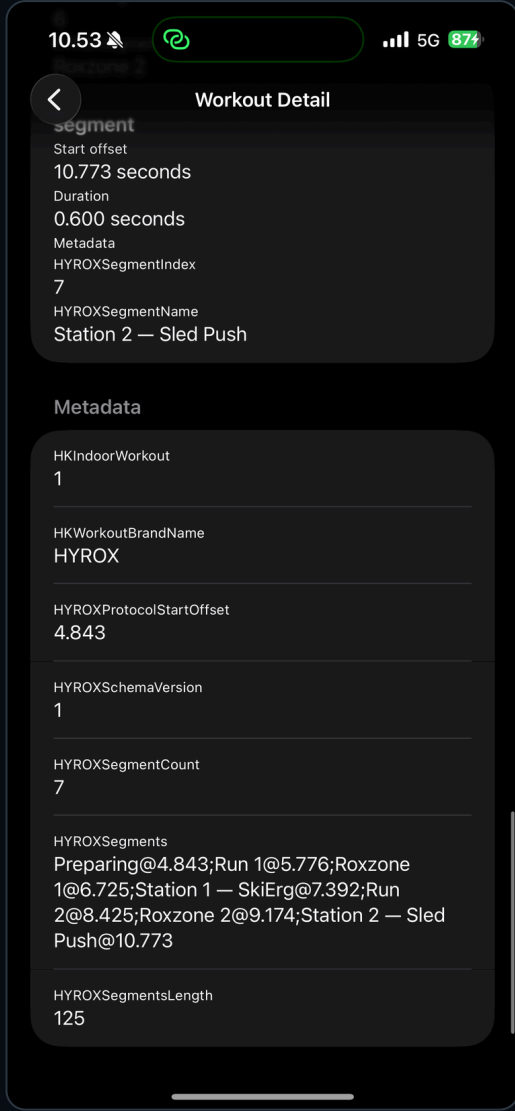
E6.2 • TWO RECORDS AGREEING

Seven events and seven names from two independent records of the same race, compared and reported with their coverage.



THE COMPARISON • APPLE’S WORKOUT

NO EVENTS – normal for apps that do not record structure. Mine records seven, and they reconcile.



E6.3 • THE ONE LINE

HKWorkoutBrandName: HYROX – the key that made Apple’s own Health app name the workout.

Two findings I could only get by breaking things.

```
CALL 1: recovered
(state: running;
startDate: 24 Aug 2026
at 9.29.49)
```

```
CALL 2: error: Task server
endpoint for 'AE606960...'
already exists (for instance
'472919D3...')
```

VERDICT: DIFFERENT

WATCH READOUT · E2.0B · TRANSCRIBED

Call 1 recovered. Call 2 failed **against call 1's own session**. Recovery is not idempotent — and that solved the mystery Cycle 1 had left open.

These two are transcribed from the watch rather than shown as screenshots — the photographs live on the device, not in this deck.

E2.2b Inspect disk

**Real file: present and
valid (1869 bytes)**

Temp file: absent

—

E2.2 Persistence

Restored Run 2; state began
36.0s ago; 4 segments
complete.

WATCH READOUT · E2.2 / E2.2B · TRANSCRIBED

Killed deliberately mid-save, at three separate points. **Nothing half-written ever survived**, and the elapsed clock carried through the interval the process did not exist for.

What exists, and where.

2 APPS ON REAL DEVICES	4,094 LINES OF SWIFT, 18 FILES	101 COMMITTS, 20 AUG – 4 SEP	2,172 WORKBOOK LINES
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The apps A watch app that records a HYROX race, survives a force quit, and writes structure as HKWorkoutEvents. A phone app that labels where every number came from. HyroxCoachWatch/ · HyroxCoach/	The workbook Every prediction before the test, every result after, every gap between – including two conclusions I had to retract. WORKBOOK.md · 26 scorecard rows	The decision records Three, including the one that says do not build the transport, with reversal conditions and accepted risks. design/L2 · L4 · L5	The evidence index All seventeen instances of the pattern, each with a workbook line reference, so the claim is checkable rather than asserted. EVIDENCE-INDEX.md	The failure log Ten bugs, each with the symptom, the layer it appeared to be in, and the layer the cause was actually in. WORKBOOK.md · running failure log
The CBL canvases Eight Miro boards: the challenge statement, the Investigate phase, and one Act board per cycle – guiding questions, findings and a synthesis on each. Miro · C4 26 · Cycles 1-6	The lo-fi design Six artboards: one watch screen, five phone screens. Correction and station identity both moved off the wrist. design/HyroxCoach-lofi.sketch			

Three clocks, one of them wrong.

During a workout the watch shows the same elapsed time computed three ways. Drop your wrist for a minute and **one of them falls behind**.

Ticks

COUNTS TIMER FIRINGS — DRIFTS

Date

TWO TIMESTAMPS — EXACT

System

RENDERED BY WATCHOS — EXACT

That one screen is my entire first cycle happening in front of you. It is also the answer to the obvious objection: **the percentages measure how often the app was allowed to run, not the accuracy of anything recorded**. Every time in the race record is a subtraction between two timestamps.

And one line of metadata: there is no HYROX workout type in HealthKit, but Apple's own Health app now displays my workout as **HYROX** — while I keep my own live screens.